Survival Analysis

Non-Parametric and Semi-Parametric Methods

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github.com/walternyamutamba-svg/rats-survival-analysis

Project Overview

Project 1: Parametric Analysis

1. Dataset & Study Design
2. Descriptive Overview
3. Kaplan–Meier Analysis
4. Nelson–Aalen Estimator
5. Hypothesis Testing
6. Parametric Model Selection
7. Weibull Life Functions
8. Frailty Model

Project 2: Extended Workflow

9. Hypothesis Testing
10. Cox Proportional Hazards
11. PH Diagnostics
12. Model Refinement
13. Synthesis & Conclusions

Objective

Objective of the Study

To determine the time to tumour development in rats and characterise the underlying hazard pattern to understand tumour onset dynamics.

Research Questions

Research Questions

- How can non-parametric and semi-parametric survival analysis methods be used to characterise the time to tumour development in rats, and what do these methods reveal about the evolution of the hazard of tumour onset and the effects of treatment and sex?
- How long do rats remain tumour-free following carcinogen exposure, and is the assumption of constant hazard over time statistically justified?

Methodology

Dataset Description

This study uses data from a tumorigenesis experiment by Mantel et al. (1977), comprising 300 rats drawn from 100 litters. Rats were randomised in a 2:1 ratio to control (200 rats) or drug treatment (100 rats), and all received a carcinogen at day 0. The outcome of interest is time to tumour development, measured in days, with observations right-censored at approximately 104 days. A total of 42 tumours were observed (event rate 14%), while 258 rats (86%) were censored. The clustering structure of the data, rats nested within litters is acknowledged as a limitation of the current analysis.

Variables

Table 1: Description of variables in the `rats` dataset.

Variable	Type	Description
`time`	Continuous	Days to tumour or censoring
`status`	Binary	**1** = tumour, **0** = censored
`rx`	Binary	1 = treated, 0 = control
`sex`	Categorical	f = female, m = male
`litter`	Categorical	Litter ID (clustering unit)

Study Design

- **100 litters** (3 rats per litter); litter serves as the clustering unit
- Randomised **2:1 allocation**: 200 control rats, 100 treated rats
- Equal representation of sex: **150 females** and **150 males**
- All rats exposed to a **carcinogen** at day 0
- Follow-up until **tumour onset** or end of study
- Right-censoring at approximately **104 days**

Kaplan-Meier: Overall Survival

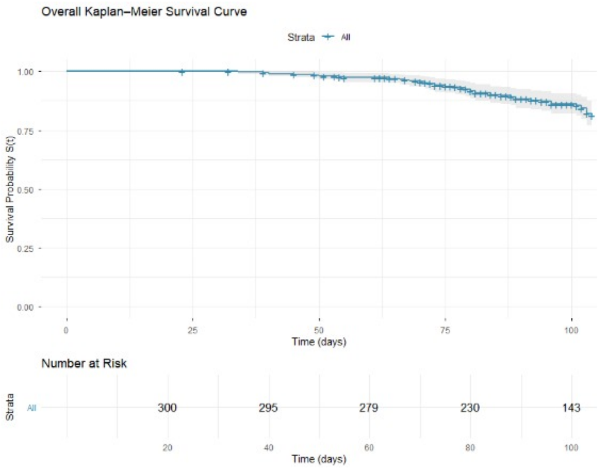

Figure 1: Overall Kaplan-Meier survival curve with 95% confidence interval.

Kaplan–Meier Analysis by Treatment Group

KM Curves by Treatment

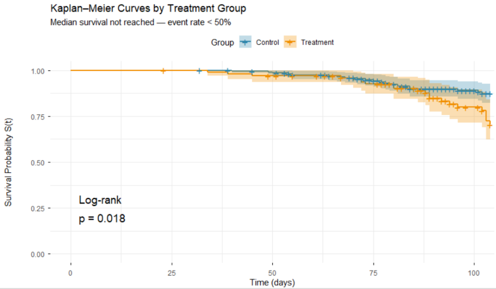

Figure 2: Kaplan–Meier survival curves by treatment group.

Median survival
Not estimable event rate $<50\%$. Curves never drop below 0.5.

Table 2: Survival probabilities at selected time points.

Time	Ctrl	Trt
25 days	1.000	1.000
50 days	0.998	0.990
75 days	0.985	0.951
100 days	0.907	0.823

Group differences
Log-rank $p = \mathbf{0.018}$ significant. Treated rats show faster decline.

Kaplan–Meier Analysis by Sex

KM Curves by Sex

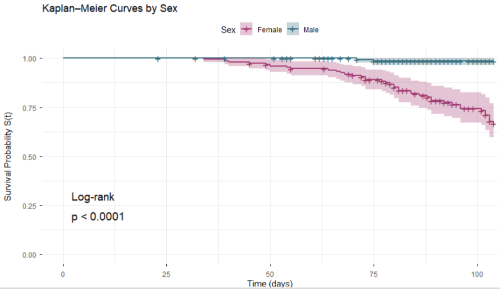

Figure 3: KM survival curves stratified by sex.

Key Findings

- Survival differs markedly by sex.
- **Female rats** show a faster decline in survival probability.
- **Male rats** maintain higher survival throughout follow-up.
- Median survival is not reached in either group due to low event rate.
- Log-rank test indicates a **statistically significant difference** between sexes.

Cumulative Hazard and Hazard Shape

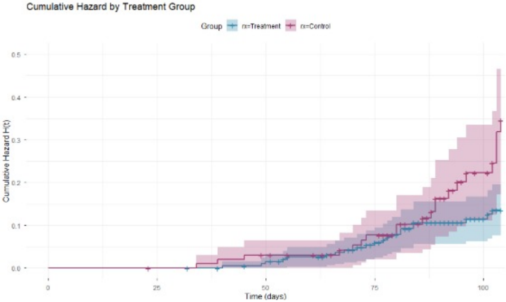

Figure 4: Nelson-Aalen cumulative hazard estimates by treatment group. Control group accumulates risk faster.

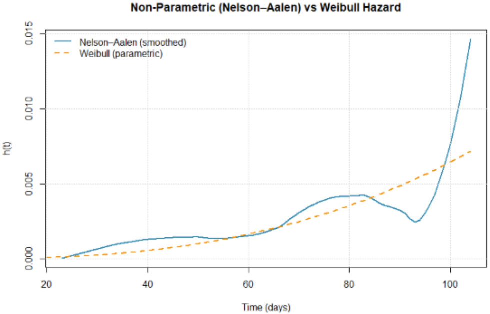

Figure 5: Smoothed hazard function. Hazard is highest early then decreases over time.

Log-Rank and Wilcoxon Test Results

Table 3: Log-Rank and Wilcoxon test results for survival differences by treatment and sex.

Comparison	**Test**	χ^2	**p-value**	**Significant?**
Treatment vs Control	Log-Rank	5.5	0.020	Yes
Treatment vs Control	Wilcoxon	5.0	0.030	Yes
Female vs Male	Log-Rank	35.9	0.000	Yes
Female vs Male	Wilcoxon	35.6	0.000	Yes

Interpretation

Both Log-Rank and Wilcoxon tests show **significant differences** in survival by treatment ($p < 0.05$) and by sex ($p < 0.000$). The agreement between both tests strengthens the conclusion.

Cox Proportional Hazards Model

Hazard Ratios for All Models

Table 4: Hazard ratios across Cox models.

Model	HR	95% CI	p
Univariable (Trt)	2.042	1.12–3.74	0.021
Adjusted (Trt)	2.206	1.20–4.04	0.011
Clustered SE	2.206	1.24–3.92	0.007
Sex (male)	0.047	0.01–0.19	<0.001

Univariable HRs
Treatment HR = 2.042 (p = 0.021). Sex (male) HR = 0.047 (p <0.001).

Adjusted effects
After adjusting for sex, sex was a partial confounder.

Tie-Handling Comparison

Table 5: Comparison of tie-handling methods.

Method	HR	p
Breslow	2.193	0.021
Efron	2.206	0.021
Exact	2.203	0.021

Influential observations
Residuals near zero; no single rat disproportionately influences estimates. Model is stable.

Linearity
Both covariates are categorical; Martingale residuals show no non-linear time trend.

PH Assumption Diagnostics

Schoenfeld Residuals

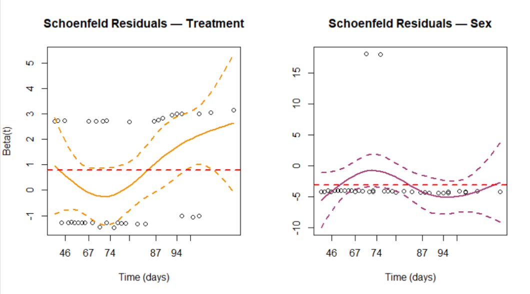

Figure 6: Schoenfeld residuals plot.

Table 6: PH assumption test results.

Variable	p	Decision
Treatment	0.020	Violated
Sex	0.438	Holds
GLOBAL	0.046	Violated

The HR between groups changes over time.

Implication

PH violation for treatment justifies:

- Stratified Cox model
- Time-dependent coefficients

Deviance Residuals and Influence

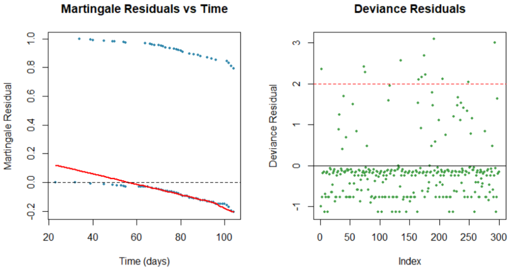

Figure 7: Deviance residuals for each observation

- **Model Fit:** Deviance residuals are randomly scattered around zero, indicating a good overall fit with no strong evidence of influential outliers.
- **Functional Form:** Martingale residuals show a slight trend over time, suggesting possible time-dependent effects, though the model captures the general pattern well.

Model Refinement

Stratified Cox (by sex)

Treatment HR estimated within each stratum.

- Treatment HR ≈ 2.2 shows consistent
- PH test after stratification: $p = 0.24$
- Lower AIC than original model

Time-Dependent Coefficient

Interaction `treatment` $\times \log(t+1)$ directly models how HR changes over time.

- Significant *tt* term confirms time-varying treatment effect
- Consistent with PH violation finding

Table 7: Model AIC comparison.

Model	AIC
Original multivariable	404.53
Stratified by sex	386.25
Time-dependent coeff.	404.61

Final model

Stratified Cox is preferred. Treatment HR remains $\approx$2.2; the PH violation does not change the direction or magnitude of the treatment effect, only its precision over time.

KM vs Parametric Model (Synthesis)

KM Curve vs Weibull Parametric Fit

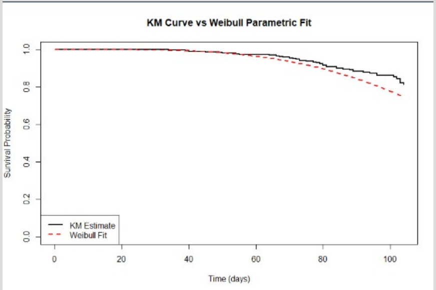

Figure 8: KM vs. Weibull survival fit.

Does Weibull capture the empirical pattern?

The Weibull survival curve tracks the KM estimate closely throughout the 104-day study. Both curves remain above 0.75, consistent with the 14% event rate.

The PP-plot from Project 1 confirmed good agreement (points near the 45° diagonal).

Nelson–Aalen vs Weibull hazard?
The smoothed NA hazard and the Weibull parametric hazard show the same monotonically increasing pattern and accelerate at the same rate after day 60.

Connecting All Methods

p-value Consistency Across Methods

Table 8: p-values across statistical methods.

Method	Trt p	Sex p
Log-rank test	0.018	<0.001
Wilcoxon test	0.023	<0.001
Univariable Cox	0.021	<0.001
Multivariable Cox	0.011	<0.001
Frailty model	0.022	—

Tests vs Cox model
All methods give treatment p $\approx$ 0.018–0.023.

Overall survival experience
Median not reached (14% events). RMST $\approx$ 99.8 days. Most rats remain tumour-free throughout.

Overall Survival Summary

Table 9: Survival summary statistics.

Quantity	Value
RMST	99.78 days
Survival at 100d (Ctrl)	90.7%
Survival at 100d (Trt)	82.3%
Hazard trend	Increasing ($k = 3.67$)

Hazard nature
Increasing hazard confirmed by both non-parametric and parametric methods. Tumour risk accelerates with age.

Key predictors
Treatment (HR = 2.21) and Sex (HR = 0.047) both significant across all models.

Weibull AFT Model

Time Ratios

Table 10: Weibull AFT time ratios.

Variable	Time Ratio
Treatment (Treated)	0.809
Sex (Male)	2.270

Interpretation:

- Treated rats develop tumours faster than controls.
- Male rats remain tumour-free longer than females.

Table 11: Cox vs. AFT model estimates.

Model	Measure	Value
Cox (adjusted)	HR Treated	2.21
Cox (frailty)	HR Treated	2.07
AFT (Weibull)	TR Treated	0.81

All three frameworks give **consistent conclusions**

- Treatment significantly accelerates tumour development
- Male sex is strongly protective
- Effect sizes are stable across methods

Limitations & Recommendations

Limitations

- **High censoring:** Many rats remained tumour-free, limiting estimation of long-term and median survival.
- **Model limitation:** The treatment effect violates the proportional hazards assumption, indicating time-varying risk.
- **Unobserved heterogeneity:** Strong litter clustering suggests additional unmeasured biological or environmental factors.

Recommendations

- **Extend follow-up:** Capture more tumour events to improve survival estimation.
- **Flexible modeling:** Use time-varying Cox or AFT models to better capture changing treatment effects.
- **Richer data:** Incorporate additional covariates and better account for litter-level variability.

Implications

- The drug shows protective effect and warrants further investigation for clinical use.
- Sex-specific differences suggest biological mechanisms influencing tumour development should be explored.
- Early high-risk period highlights the importance of early intervention after carcinogen exposure.

Conclusion

- Treatment is associated with a significant increase in tumour risk.
- Sex is the strongest determinant, with males showing substantially lower risk than females.
- Tumour risk increases over time, supporting a non-constant hazard.
- There is strong clustering within litters, indicating important underlying heterogeneity.
- Overall, results are consistent across all modelling approaches, strengthening confidence in the findings.

References

Mantel et al. (1977), *Cancer Research*.

Therneau & Grambsch (2000), *Springer*.

Thank You

Questions?

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